

No books, notes, computational devices, etc. are allowed. Use clear handwriting and give clear careful motivations. Please write your anonymous identification number on each sheet of paper.

1. Let G be a finite group with an even number of conjugacy classes. Prove that $|G|$ is even.
2. Let G be a group of order pqr , for distinct primes $p < q < r$. Show that G has a normal Sylow subgroup, and prove further that G is solvable.
3. The set $\mathbb{C}^*$ of non-zero complex numbers is an abelian group under multiplication, and thus is also a $\mathbb{Z}$ -module under the action $r \cdot x = x^r$ for $x \in \mathbb{C}^*$ and $r \in \mathbb{Z}$. Let p be a prime, and consider the subset

$$U = \{x \in \mathbb{C}^* \mid x^{p^n} = 1 \text{ for some } n\}.$$

Show that U is a $\mathbb{Z}$ -submodule.

Show further that all proper submodules of U are of the form $U_m = \{x \in \mathbb{C}^* \mid x^{p^m} = 1\}$ for some $m \in \mathbb{N}$.

Is U a Noetherian $\mathbb{Z}$ -module? Is U an Artinian $\mathbb{Z}$ -module?

4. Find all $\mathbb{Z}$ -homomorphisms from $\mathbb{Z}/30\mathbb{Z}$ to $\mathbb{Z}/12\mathbb{Z}$.
5. Find the invariant factors of the following matrix over $\mathbb{Q}[x]$:

$$\begin{pmatrix} 1-x & 2 & -4 & 4 \\ 2 & -1-x & 4 & -8 \\ 1 & 0 & 1-x & -2 \\ 0 & 1 & -2 & 3-x \end{pmatrix}$$

6. Determine the abelian group structure of $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/3\mathbb{Z}$ and of $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$.